

SALES ORDERS DATASET

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1. COMMIT;

```
mysql> START TRANSACTION;
Query OK, 0 rows affected (0.00 sec)

mysql>
mysql> INSERT INTO sales_orders (
  ->   order_id,
  ->   order_date,
  ->   customer_name,
  ->   product_category,
  ->   product_name,
  ->   quantity,
  ->   unit_price,
  ->   total_amount,
  ->   payment_method,
  ->   region
  -> )
  -> VALUES (
  ->   2001,
  ->   '2025-01-15',
  ->   'John Doe',
  ->   'Electronics',
  ->   'Headphones',
  ->   5,
  ->   300.00,
  ->   1500.00,
  ->   'Credit Card',
  ->   'North America'
  -> );
Query OK, 1 row affected (0.00 sec)

mysql>
mysql> COMMIT;
Query OK, 0 rows affected (0.00 sec)

mysql> SELECT * FROM sales_orders WHERE order_id = 2001;
+-----+-----+-----+-----+-----+-----+-----+-----+-----+-----+
| order_id | order_date | customer_name | product_category | product_name | quantity | unit_price | total_amount | payment_method | region |
+-----+-----+-----+-----+-----+-----+-----+-----+-----+-----+
|      2001 | 2025-01-15 | John Doe     | Electronics     | Headphones   |         5 |      300.00 |      1500.00 | Credit Card   | North America |
+-----+-----+-----+-----+-----+-----+-----+-----+-----+-----+
1 row in set (0.01 sec)

mysql>
```

2. ROLLBACK;

```
mysql> START TRANSACTION;
Query OK, 0 rows affected (0.00 sec)

mysql>
mysql> UPDATE sales_orders
  -> SET quantity = quantity - 100
  -> WHERE order_id = 1;
Query OK, 0 rows affected (0.01 sec)
Rows matched: 0  Changed: 0  Warnings: 0

mysql>
mysql> ROLLBACK;
Query OK, 0 rows affected (0.00 sec)
```

3. SAVEPOINT;

```
mysql> START TRANSACTION;  
Query OK, 0 rows affected (0.00 sec)
```

```
mysql>  
mysql> UPDATE sales_orders  
-> SET quantity = quantity - 10  
-> WHERE order_id = 2001;  
Query OK, 1 row affected (0.00 sec)  
Rows matched: 1 Changed: 1 Warnings: 0
```

```
mysql>  
mysql> SAVEPOINT stock_updated;  
Query OK, 0 rows affected (0.00 sec)
```

```
mysql>  
mysql> INSERT INTO sales_orders (  
-> order_id,  
-> order_date,  
-> customer_name,  
-> product_category,  
-> product_name,  
-> quantity,  
-> unit_price,  
-> total_amount,  
-> payment_method,  
-> region  
-> )  
-> VALUES (  
-> 99999,  
-> '2025-01-15',  
-> 'Jane Doe',  
-> 'Electronics',  
-> 'Headphones',  
-> 5,  
-> 300.00,  
-> 1500.00,  
-> 'Failed Payment',  
-> 'North America'  
-> );  
Query OK, 1 row affected (0.00 sec)
```

```
mysql>  
mysql> ROLLBACK TO SAVEPOINT stock_updated;  
Query OK, 0 rows affected (0.00 sec)
```

```
mysql>  
mysql> COMMIT;  
Query OK, 0 rows affected (0.00 sec)
```

```
mysql> SELECT order_id, quantity FROM sales_orders WHERE order_id = 2001;  
+-----+-----+  
| order_id | quantity |  
+-----+-----+  
| 2001 | -5 |  
+-----+-----+  
1 row in set (0.00 sec)
```

```
mysql> SELECT * FROM sales_orders WHERE order_id = 99999;  
Empty set (0.00 sec)
```

4. ROLLBACK TO SAVEPOINT

```
mysql> START TRANSACTION;
Query OK, 0 rows affected (0.00 sec)

mysql>
mysql> UPDATE sales_orders
  -> SET total_amount = total_amount + 50
  -> WHERE order_id = 2001;
Query OK, 1 row affected (0.01 sec)
Rows matched: 1  Changed: 1  Warnings: 0

mysql>
mysql> SAVEPOINT points_added;
Query OK, 0 rows affected (0.00 sec)

mysql>
mysql> DELETE FROM sales_orders
  -> WHERE order_id = 99999;
Query OK, 0 rows affected (0.00 sec)

mysql>
mysql> ROLLBACK TO SAVEPOINT points_added;
Query OK, 0 rows affected (0.00 sec)

mysql>
mysql> COMMIT;
Query OK, 0 rows affected (0.01 sec)
```

```
mysql> SELECT order_id, total_amount FROM sales_orders WHERE order_id = 2001;
+-----+-----+
| order_id | total_amount |
+-----+-----+
|      2001 |       1550.00 |
+-----+-----+
1 row in set (0.00 sec)

mysql> SELECT order_id FROM sales_orders WHERE order_id = 99999;
Empty set (0.00 sec)
```

5. RELEASE SAVEPOINT

```
mysql> START TRANSACTION;
Query OK, 0 rows affected (0.00 sec)

mysql>
mysql> UPDATE sales_orders
  -> SET unit_price = unit_price * 0.9
  -> WHERE product_category = 'Electronics';
Query OK, 1 row affected (0.01 sec)
Rows matched: 1  Changed: 1  Warnings: 0

mysql>
mysql> SAVEPOINT discount_applied;
Query OK, 0 rows affected (0.00 sec)

mysql>
mysql> RELEASE SAVEPOINT discount_applied;
Query OK, 0 rows affected (0.00 sec)

mysql>
mysql> COMMIT;
Query OK, 0 rows affected (0.00 sec)

mysql> _
```

```
mysql> SELECT order_id, product_name, product_category, unit_price from sales_orders where product_category = "Electronics";
+-----+-----+-----+-----+
| order_id | product_name | product_category | unit_price |
+-----+-----+-----+-----+
|      2001 | Headphones   | Electronics      |      270.00 |
+-----+-----+-----+-----+
1 row in set (0.00 sec)
```

6. SET TRANSACTION;

```
mysql> SET TRANSACTION ISOLATION LEVEL READ COMMITTED;
Query OK, 0 rows affected (0.01 sec)

mysql>
mysql> START TRANSACTION;
Query OK, 0 rows affected (0.00 sec)

mysql>
mysql> SELECT * FROM sales_orders WHERE order_id = 2001;
+-----+-----+-----+-----+-----+-----+-----+-----+-----+
| order_id | order_date | customer_name | product_category | product_name | quantity | unit_price | total_amount | payment_method | region |
+-----+-----+-----+-----+-----+-----+-----+-----+-----+
|      2001 | 2025-01-15 | John Doe     | Electronics      | Headphones   |      -5 |      270.00 |      1550.00 | Credit Card   | North America |
+-----+-----+-----+-----+-----+-----+-----+-----+-----+
1 row in set (0.00 sec)

mysql>
mysql> COMMIT;
Query OK, 0 rows affected (0.00 sec)
```

7. ORDER PLACEMENT TRANSACTION

```
mysql> START TRANSACTION;
Query OK, 0 rows affected (0.00 sec)

mysql>
mysql> UPDATE sales_orders
  -> SET quantity = quantity - 3
  -> WHERE order_id = 2001;
Query OK, 1 row affected (0.00 sec)
Rows matched: 1  Changed: 1  Warnings: 0

mysql>
mysql> INSERT INTO sales_orders (
  ->   order_id,
  ->   order_date,
  ->   customer_name,
  ->   product_category,
  ->   product_name,
  ->   quantity,
  ->   unit_price,
  ->   total_amount,
  ->   payment_method,
  ->   region
  -> )
  -> VALUES (
  ->   5002,
  ->   '2025-02-01',
  ->   'Customer 2002',
  ->   'Electronics',
  ->   'Smartphone',
  ->   3,
  ->   299.67,
  ->   899.00,
  ->   'SUCCESS',
  ->   'North America'
  -> );
ERROR 1062 (23000): Duplicate entry '5002' for key 'PRIMARY'
mysql>
mysql> COMMIT;
Query OK, 0 rows affected (0.00 sec)
```

```
mysql> select order_id, product_name, quantity from sales_orders where order_id = 2001;
+-----+-----+-----+
| order_id | product_name | quantity |
+-----+-----+-----+
|      2001 | Headphones   |        -8 |
+-----+-----+-----+
1 row in set (0.00 sec)
```

```
mysql> select order_id, order_date, total_amount, payment_method from sales_orders where order_id = 5002;
+-----+-----+-----+-----+-----+
| order_id | order_date | total_amount | payment_method |
+-----+-----+-----+-----+-----+
|      5002 | 2025-02-01 |         899.00 | SUCCESS        |
+-----+-----+-----+-----+-----+
1 row in set (0.00 sec)
```

8. REFUND/CANCELLATION WITH ROLLBACK SAFETY

```
mysql> START TRANSACTION;
Query OK, 0 rows affected (0.00 sec)

mysql>
mysql> UPDATE sales_orders
  -> SET quantity = quantity + 3
  -> WHERE order_id = 2001;
Query OK, 1 row affected (0.00 sec)
Rows matched: 1  Changed: 1  Warnings: 0

mysql>
mysql> UPDATE sales_orders
  -> SET payment_method = 'CANCELLED'
  -> WHERE order_id = 5002;
Query OK, 1 row affected (0.00 sec)
Rows matched: 1  Changed: 1  Warnings: 0

mysql>
mysql> SAVEPOINT refund_point;
Query OK, 0 rows affected (0.00 sec)

mysql>
mysql> UPDATE sales_orders
  -> SET payment_method = 'REFUND_FAILED'
  -> WHERE order_id = 5002;
Query OK, 1 row affected (0.00 sec)
Rows matched: 1  Changed: 1  Warnings: 0

mysql>
mysql> ROLLBACK TO SAVEPOINT refund_point;
Query OK, 0 rows affected (0.00 sec)

mysql>
mysql> COMMIT;
Query OK, 0 rows affected (0.01 sec)
```

```
mysql> select order_id, quantity from sales_orders where order_id = 2001;
+-----+-----+
| order_id | quantity |
+-----+-----+
|      2001 |        -5 |
+-----+-----+
1 row in set (0.00 sec)
```

```
mysql> select order_id, payment_method, from sales_orders where order_id = 5002;
ERROR 1064 (42000): You have an error in your SQL syntax; check the manual that corresponds to
mysql> select order_id, payment_method from sales_orders where order_id = 5002;
+-----+-----+
| order_id | payment_method |
+-----+-----+
|      5002 | CANCELLED      |
+-----+-----+
1 row in set (0.00 sec)
```

9. BULK PRICE UPDATE WITH SAFETY NET

```
mysql>
mysql> START TRANSACTION;
Query OK, 0 rows affected (0.00 sec)

mysql>
mysql> SAVEPOINT before_price_update;
Query OK, 0 rows affected (0.00 sec)

mysql>
mysql> UPDATE sales_orders
      -> SET unit_price = unit_price * 1.1
      -> WHERE product_category = 'Electronics';
Query OK, 2 rows affected, 1 warning (0.00 sec)
Rows matched: 2  Changed: 2  Warnings: 1

mysql>
mysql> ROLLBACK TO SAVEPOINT before_price_update;
Query OK, 0 rows affected (0.01 sec)

mysql>
mysql> COMMIT;
Query OK, 0 rows affected (0.00 sec)

mysql> select order_id, product_name, product_category, unit_price from sales_orders where product_category = "Electronics";
+-----+-----+-----+-----+
| order_id | product_name | product_category | unit_price |
+-----+-----+-----+-----+
|      2001 | Headphones   | Electronics      |      270.00 |
|      5002 | Smartphone   | Electronics      |      299.67 |
+-----+-----+-----+-----+
2 rows in set (0.00 sec)
```

10. MULTI STEP CHECKOUT WITH PARTIAL ROLLBACK;

```
mysql> START TRANSACTION;
Query OK, 0 rows affected (0.00 sec)

mysql>
mysql> UPDATE sales_orders
      -> SET quantity = quantity - 1
      -> WHERE order_id = 2001;
Query OK, 1 row affected (0.00 sec)
Rows matched: 1  Changed: 1  Warnings: 0

mysql>
mysql> SAVEPOINT item1_reserved;
Query OK, 0 rows affected (0.00 sec)

mysql>
mysql> UPDATE sales_orders
      -> SET quantity = quantity - 1
      -> WHERE order_id = 5002;
Query OK, 1 row affected (0.00 sec)
Rows matched: 1  Changed: 1  Warnings: 0

mysql>
mysql> ROLLBACK TO SAVEPOINT item1_reserved;
Query OK, 0 rows affected (0.00 sec)

mysql>
mysql> COMMIT;
Query OK, 0 rows affected (0.00 sec)
```

```
mysql> select order_id, product_name, quantity, from sales_orders where order_id = 2001;
ERROR 1064 (42000): You have an error in your SQL syntax; check the manual that corresponds to your My
mysql> select order_id, product_name, quantity from sales_orders where order_id = 2001;
+-----+-----+-----+
| order_id | product_name | quantity |
+-----+-----+-----+
|      2001 | Headphones   |        -6 |
+-----+-----+-----+
1 row in set (0.00 sec)

mysql> select order_id, product_name, quantity from sales_orders where order_id = ;
ERROR 1064 (42000): You have an error in your SQL syntax; check the manual that corresponds to your My
mysql> select order_id, product_name, quantity from sales_orders where order_id = 5002;
+-----+-----+-----+
| order_id | product_name | quantity |
+-----+-----+-----+
|      5002 | Smartphone   |         3 |
+-----+-----+-----+
1 row in set (0.00 sec)
```